

DATA SCIENCE RESOURCES

Time Series Forecasting Projects

Project 1 : Stock Price Prediction

Problem Statement: Investors and financial institutions require reliable models to forecast stock prices for decision-making, risk management, and portfolio optimization.

Objective: Build a model that predicts future stock prices using historical price trends and market indicators.

Dataset Source

- [Yahoo Finance API using the yfinance Python library](#)
- [NSE/BSE historical data from Kaggle or Quandl](#)

Techniques and Concepts

- Time series analysis and decomposition
- Moving averages, RSI, Bollinger Bands
- ARIMA, SARIMA, Prophet
- LSTM or GRU for deep learning time series models

Tools and Libraries: Python, yfinance, pandas, matplotlib, statsmodels, scikit-learn, Keras, TensorFlow

Expected Outcome: A model that accurately predicts the next-day or next-week closing price of a selected stock, supported by visual plots and metrics like RMSE or MAE.

Project 2 : Electricity Demand Forecasting

Problem Statement: Energy providers must predict electricity consumption in advance to manage power generation, avoid blackouts, and optimize grid operations.

Objective: Create a model to forecast hourly or daily energy demand using historical data and seasonal patterns.

Dataset Source

- [New York ISO Hourly Energy Consumption \(Kaggle\)](#)
- [Australian Smart Grid Dataset \(UCI ML Repository\)](#)

Techniques and Concepts

- Seasonal decomposition
- SARIMA, Prophet
- Feature engineering with lags and rolling statistics
- Gradient boosting with temporal features

Tools and Libraries: Python, pandas, Prophet, XGBoost, LightGBM, matplotlib

Expected Outcome: A model that forecasts electricity usage in a given region with time-based visualization and performance metrics.

Project 3 : COVID-19 Daily Cases Forecast

Problem Statement: Accurate prediction of daily COVID-19 cases is essential for health planning, hospital resource allocation, and containment strategies.

Objective: Forecast daily confirmed COVID-19 cases based on historical trends using time series techniques.

Dataset Source

- [Johns Hopkins COVID-19 Repository](#)
- [COVID-19 India Dataset via public API](#)

Techniques and Concepts

- Rolling averages and smoothing
- Curve fitting with Prophet
- LSTM for long-term sequential prediction
- ARIMA and exponential smoothing

Tools and Libraries: Python, pandas, Prophet, Keras, matplotlib

Expected Outcome: A forecast of daily new cases over the next 7 to 30 days with uncertainty intervals and performance visualizations.

Project 4 : Retail Sales Forecasting

Problem Statement: Retailers need to accurately predict future sales to optimize inventory, reduce overstock or stockouts, and improve revenue forecasting.

Objective: Develop a model that forecasts future sales at the product or store level using historical data and calendar variables.

Dataset Source

- [Rossmann Store Sales Dataset \(Kaggle\)](#)
- [Walmart Store Sales Forecasting Competition](#)

Techniques and Concepts

- Time series decomposition
- Feature engineering (holiday, store type, promo)
- SARIMA and Prophet for time series modeling
- XGBoost/LightGBM with temporal data

Tools and Libraries: Python, pandas, Prophet, XGBoost, LightGBM, matplotlib, seaborn

Expected Outcome: A prediction system that estimates store or item sales for the next weeks or months, including holidays and promotions.
